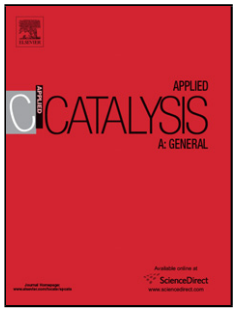
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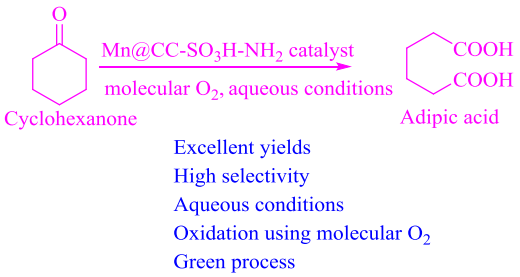
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Manganese incorporated on glucose as an efficient catalyst for the synthesis of adipic acid using molecular O_2 in aqueous medium

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Graphical abstract



Highlights

- A new manganese incorporated on glucose catalyst Mn@CC-SO3H-NH2was synthesized.
- Selective synthesis of adipic acid starting from cyclohexanone using molecular O2 as a sole oxidant under aqueous condition gave 82% yield with 100% selectivity.
- The Mn@CC-SO3H-NH2 catalyst was found to be highly efficient, highly selective, easily separable, reusable and environment friendly.

ABSTRACT

A new manganese incorporated on glucose catalyst Mn@CC-SO_3H-NH_2 was synthesized and well characterized using FT-IR, PXRD, EDAX, SEM, TEM, ^{13}C cross-polarization NMR and BET analysis. The catalyst thus synthesized was studied for its utility in the synthesis of industrially important chemical adipic acid starting from readily available cyclohexanone using molecular O_2 as a source. Adipic acid was obtained in maximum yield of 82% using Mn@CC-SO_3H-NH_2 catalyst in water as a solvent. The catalyst was easily recovered and reused for minimum five runs without any significant loss in the yield of products. Using molecular O_2 as an oxidant and water as a solvent, the reaction is very important from green chemistry perspectives and can also be industrially feasible.

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Keywords: Manganese, catalyst, adipic acid, water, recyclable.

1. Introduction

Adipic acid is one of the most important industrial chemical required all over the world.[1] The worldwide industrial production of adipic acid is 2.2 million metric tons per year. It is considered to be the building block for number of different products that are manufactured in chemical, pharmaceutical and food industries. It is the main precursor to Nylon-6,6 and also used for the production of polyester, polyresin, polyvinyl chloride.[2] Adipic acid is also used in cosmetics, fertilizers, paper and wax industry. At present adipic acid is produced industrially by catalytic oxidation of a mixture of cyclohexanol and cyclohexanone using nitric acid.[3] The major drawback for this process is the emission of harmful nitrous oxide N_2O that is commonly thought to cause global warming and ozone depletion as well as acid rain and smog.[4] Adipic acid producers have to face challenge to decompose this N_2O safely without causing any impact on the pollution.

Efforts have been made to develop process for synthesis of adipic acid that can overcome conventional process. Synthesis starting from cyclohexanone, cyclohexanol, cyclohexene and cyclohexane have been explored. Bohstrom *et al.* carried out synthesis of cyclohexene to adipic acid using tungsten oxide as a catalyst and H_2O_2 as an oxidant.[5] Chavan *et al*. used Co/Mn catalyst in acetic acid to get 86% selectivity of adipic acid from cyclohexanone.[6] Sato *et al*. obtained 78 % yield of adipic acid from cyclohexanol using Na_2WO_4 catalyst & 30% H_2O_2.[7] Lee *et al*. used titanium framework substituted aluminophosphate (TAPO-5) as a catalyst for oxidation of cyclohexene to adipic acid using

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H_2O_2 as an oxidant.[8] Several other tungstate based catalyst have been reported as well e.g. $H_3PW_{12}O_{40}$,[9] $[WO(O_2)_2.2QOH]$,[10] $[BMIm]_2WO_4$ supported on silica sulphamic acid,[11] oxotungsten containing SBA-15,[12] surfactant peroxotungstates.[13] Although reports are available in literature for synthesis of adipic acid,[14-22] the cost of oxidants like H_2O_2, *t*-butylhydroperoxide, drastic reaction conditions, using organic solvents, low yields and selectivity of adipic acid restricts its use on industrial scale. Thus there still remains a challenge for researchers to develop an efficient and stable catalyst to develop an industrially scalable process for the production of adipic acid.

2. Experimental

General Methods. All commercially available reagents were used without further purification unless otherwise stated. All reactions were carried out using flame-dried glassware with magnetic stirring. The Mn@CC-SO_3H-NH_2 catalyst thus obtained was characterized by known techniques such as FT-IR (Spectrum 400), PXRD (Panalytical X'Pert Pro), Elemental analysis from EDAX (Nova Nano SEM 450), SEM (Quanta™ Scanning Electron Microscope), TEM (Tecnai 4220 S-Twin), ^{13}C cross-polarization (CP) magic angle spining NMR (Jeol ECX 400, at a frequency of 100 MHz with a spinning rate of 8 KHz) and BET surface area (Micrometrices ASAP 2020). ^1H and ^{13}C NMR spectra were recorded on a Bruker spectrometer using TMS as an internal standard. NMR data are reported as follows: chemical shift, multiplicity (br = broad singlet, s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet), coupling constant (Hz) and integration.

Typical procedure for Mn@CC-SO_3H-NH_2 catalyst preparation

Glucose (2 g) and *p*-toluene sulfonic acid (0.4 g) were taken in a clean and dry round bottom flask and stirred at 180 °C for 24 h under nitrogen. After 24 h, a black solid mass was obtained which was then washed with millipore water and then with ethanol to obtain a black

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powder. This black powder was then dried in the oven at 100 ºC for 5 h. After drying to this black powder was added 10 mL of ethanol and 3-amino propyl triethoxy silane (APTES) (0.24 mL), sonicated for 30 min and further refluxed at 80 °C for 8 h. The solid was collected by filtration and washed with water followed by ethanol and dried in the oven at 100 ºC for 5 h. This black powder was denoted as CC-SO_3H-NH_2. After drying, to this black powder (1 g) (CC-SO_3H-NH_2) was added 10 mL ethanol, $MnCl_2$ (2 g) and further refluxed 80 °C for 12 h. The solid was collected by filtration and washed with water followed by ethanol and dried in oven at 100 °C for 5-6 h. This black powder was denoted as Mn@CC-SO_3H-NH_2 catalyst.

General Procedure for synthesis of adipic acid from cyclohexanone by using Mn@CC-SO_3H-NH_2 catalysts

A 50 mL two neck round bottom flask was equipped with a reflux condenser. The flask was charged with 0.1 mL of cyclohexanone, 5 mL of water, 20 wt % of Mn@CC-SO_3H-NH_2 catalyst. Oxygen was passed through the reaction at a flow of 20 mL/min measured using a flowmeter. The reaction mixture was then refluxed for required time and the reaction was monitored using TLC. After completion of the reaction, the reaction mixture was cooled to room temperature. The reaction mixture was filtered to separate the catalyst. The catalyst was washed 3-4 times with water followed by methanol. The filtrate was concentrated under vacuum. The residue left after concentration was washed with ethyl acetate 2-3 times to remove the unreacted cyclohexanone leaving behind colourless white solid of crude adipic acid.

3. Results and Discussion

Biomass derived carbonaceous catalyst are promising as biomass is renewable, abundant, cheap and easy for preparation. Carbonaceous catalyst consists of carbon skeleton that is stable, insoluble in most acidic/basic conditions as well as organic solvents and easily separable. Due to our continuous interest in the field of carbonaceous solid acid catalyst,[23-25]

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we were interested in the synthesis of Mn@CC-SO_3H-NH_2 catalyst and showed its remarkable catalytic activity in synthesis of one of the important value added chemical adipic acid using molecular oxygen as a sole oxidant under mild reaction conditions. For the preparation of Mn@CC-SO_3H-NH_2 catalyst, D-glucose was used as a carbon precursor. Hydrothermal carbonization of glucose and its sulfonation was done in one step by heating glucose in the presence of *p*-TSA at 180 ºC. The hydrothermal carbonization promoted the structure to undergo decomposition, hydrolysis, dehydration, and molecular rearrangement reaction to produce a polycyclic aromatic structure embedded with active -SO_3H, -COOH and -OH sites.[26] In the second step, 3-amino propyl triethoxy silane in ethanol was added to generate the basic -*O*Si-$CH_2CH_2CH_2NH_2$ site on the catalyst followed by incorporation of manganese on the catalyst surface using manganese chloride in ethanol.

The Mn@CC-SO_3H-NH_2 catalyst thus synthesized was used in the synthesis of adipic acid from cyclohexanone. The FT-IR spectrum (Figure S1) showed peaks for O=S=O stretching vibrations in -SO_3H groups at 1009 cm^{-1} and 1031 cm^{-1} and peak for SO_3- stretching at 1154 cm^{-1}, indicating the presence of sulfonic acid groups on the catalyst surface. Peak at 1619 cm^{-1} and 1702 cm^{-1} attributed to the C=C stretching vibrations in aromatic carbons and C=O stretching vibration of -COOH group. Doublet at 2926 cm^{-1} attributed for C-H stretching band and a peak at 1440 cm^{-1} indicated C-H bending vibration of alkyl groups of amine terminal silanes. Stretching vibrations for Si-O at 1095 cm^{-1} and stretching and bending vibrations of Si-O-C at 1122 cm^{-1} and 680 cm^{-1} indicated that silylanization occurred on the catalyst surface. The peak at 1591 cm^{-1} due to the N-H bending vibrations confirmed the presence of amino group on the surface of the catalyst. Bands due to O-H stretching were observed at 3429 cm^{-1}. The powder X ray diffraction pattern (Figure 2a) of Mn@CC-SO_3H-NH_2 showed broad diffraction peak at 2θ = 15 to 30º, which can be attributed to amorphous

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carbon composed of aromatic carbon sheets oriented in considerably random fashion. The presence of nitrogen, sulphur, silicon, manganese and chloride was confirmed by Energy dispersive X-ray analysis (EDAX) (Figure S2), thus confirming the acidic -SO_3H, basic *O*Si-$CH_2CH_2CH_2NH_2$ and coordination of manganese to the NH groups on the surface of the Mn@CC-SO_3H-NH_2 catalyst. SEM analysis (Figure 2b) showed a typical planar structure of Mn@CC-SO_3H-NH_2 carbonaceous catalyst maintaining two dimensional structures with heavy crumpling features. TEM analysis (Figure 2c) showed porous nature of Mn@CC-SO_3H-NH_2 carbonaceous catalyst. ^{13}C CP/MAS NMR spectrum (Figure 2d) showed peaks at 128, 154 and 174 ppm indicating presence of polycyclic aromatic carbon atoms, phenolic -OH and -COOH groups respectively. Broad peak was observed at 140 ppm indicating aromatic carbons attached to -SO_3H. Peaks from 12 ppm to 61 ppm showed the presence of *N*-propyl carbons. Using the standard Brunauer-Emmett-Teller (BET) equation BET surface areas and pore size were found to be 1.93 m^2/g and 8.96 A°.

The performance of Mn@CC-SO_3H-NH_2 catalyst was studied for the synthesis of adipic acid starting from readily available cyclohexanone. Initially the formation of adipic acid was carried out in various solvents using Mn@CC-SO_3H-NH_2 catalyst at given temperatures using molecular O_2 (20 mL/min) (Table 1). Solvents such as water, DMF, DMSO, acetonitrile, isopropanol and solvent free conditions were studied for adipic acid formation. It was observed that in DMF, DMSO and isopropanol the starting remained unconsumed (Table 1, entries 2, 3 & 5) whereas in water, acetonitrile and solvent free conditions a yield of about 81%, 82% and 80 % of adipic acid was obtained respectively.

After completion of the reaction, the solvent was removed under reduced pressure. The unreacted cyclohexanone was removed by adding ethyl acetate to the remaining crude mass. The ethyl acetate layer was decanted leaving behind white crystalline solid of adipic acid. The ^1H NMR of this white crystalline solid showed selective formation of adipic acid with

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100% purity and selectivity (Figure S3). The unreacted cyclohexanone that remained during the reaction was easily recovered and could be reused in further reactions (Figure S4).

Considering the principles of green chemistry, we optimized the reaction conditions for synthesis of adipic acid in water as a solvent and also under solvent free conditions. The reaction was studied at different temperature i.e. 100 ºC, 120 ºC and 140 ºC in water (Table 2, entries 1-3) and under solvent free conditions (Table 2, entries 4-6). It was observed that in water and solvent free conditions maximum yield was obtained at 120 ºC with less amount of cyclohexanone remaining unreacted. As the yields of adipic acid were similar both in case of water and solvent free conditions, we optimized the further reactions for adipic acid synthesis in water as a solvent, thus the process can be considered as economic, green and environment friendly.

Catalyst concentration for Mn@CC-SO_3H-NH_2 was studied for the synthesis of adipic acid in water (Table 3). It was observed that as the catalyst concentration increased from 5 wt% to 20 wt% the yield of adipic acid was increased from 62% to 82% respectively. Further increasing the catalyst concentration did not have any effect on yield of adipic acid.

Comparison of Mn@CC-SO_3H-NH_2 catalyst *vs*. literature known catalyst for synthesis of adipic acid have been shown in Table 4. Catalyst such as hybrid iron phosphonate,[14] Titano silicate molecular sieves,[17] hollow structured TS-1,[17] Mn modified hollow structured TS-1[17] did not give 100% selectivity of adipic acid instead by products like glutaric acid, succinic acid were formed (Table 4, entries 1-4). Hybrid tin phosphate,[20] mesoporous triphosphate[20] and H_2WO_4[21] gave about 74%, 22% and 91% yield of adipic acid (Table 4, entries 5-7). In this work, Mn@CC-SO_3H-NH_2 catalyst (Table 4, entry 8) was found to give 82 % yield of adipic acid with 100% selectivity which indicating good efficiency of the catalyst over known literature catalyst for selective synthesis of adipic acid.

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As the catalyst was easily recoverable at the end of the reaction by simple filtration, the catalyst was studied for its reusability in the synthesis of adipic acid. As shown in Table 5, Mn@CC-SO_3H-NH_2 catalyst exhibited consistent yield of the adipic acid even after subsequent five runs and also the unreacted cyclohexanone which was isolated can be collected and reused. Synthesis of adipic acid from cyclohexanone was also done on higher scale. When the reaction was performed on a scale of 10 g of cyclohexanone using 20 wt% Mn@CC-SO_3H-NH_2 catalyst and O_2 (20 mL/min) in water at 120 ºC after 30 h, it was observed that 82% yield of adipic acid was obtained and the unreacted cyclohexanone could be recovered and reused.

4. Conclusion

We have developed an efficient approach for the synthesis of industrially important chemical adipic acid using the newly synthesized Mn@CC-SO_3H-NH_2 catalyst using O_2 as an oxidant in water as a solvent. Adipic acid was isolated in good yield and the unreacted cyclohexanone can be easily separated and reused again for the synthesis of adipic acid. The catalyst could also be easily recovered and reused without any significant loss in the yield of

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product. The reaction conditions are economic and environmentally green compared to the present industrial procedure available for adipic acid synthesis.

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Supplementary data

Supplementary data i.e. spectral data associated with this article can be found, in the online version.

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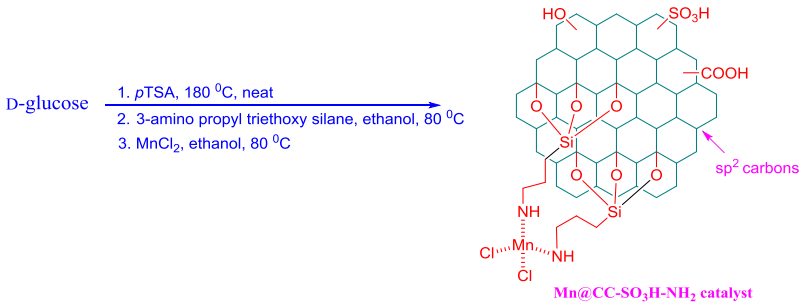


Figure 1. Preparation of Mn@CC-SO_3H-NH_2 catalyst

Fig 1

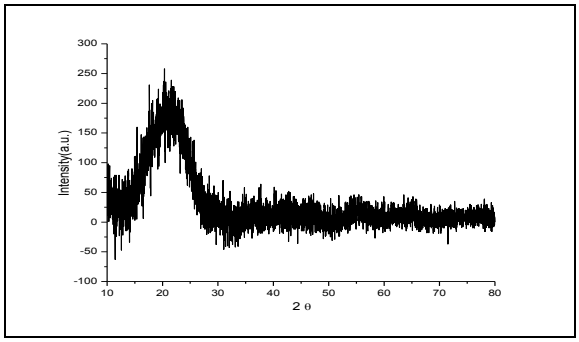


Fig 2a. XRD spectrum of Mn@CC-SO_3H-NH_2

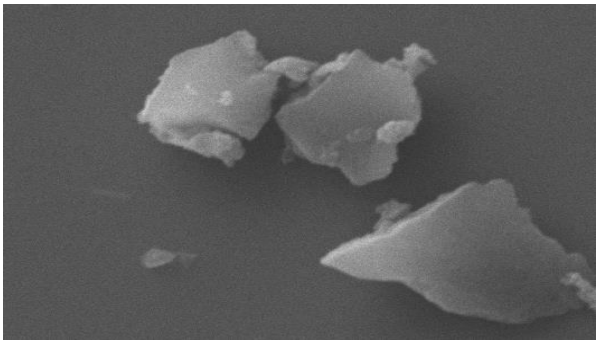


Fig 2b. SEM image of Mn@CC-SO_3H-NH_2

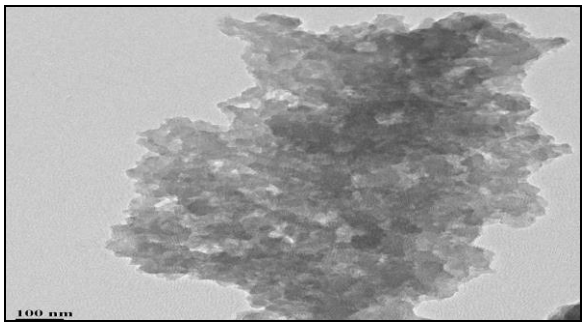


Fig 2c. TEM image of Mn@CC-SO_3H-NH_2

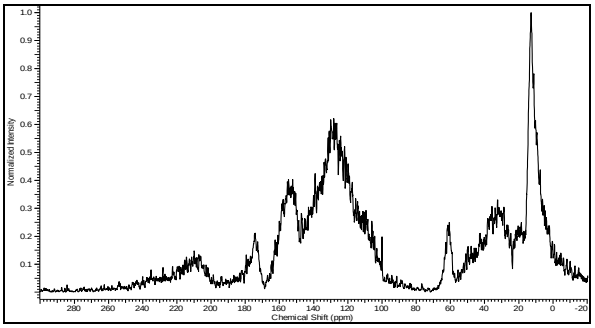


Fig 2d.^{13}C CP/MAS spectrum of Mn@CC-SO_3H-NH_2

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Table 1. Study of adipic acid synthesis in various solvents.[a]

Entry	**Solvent**	**Temp (°C)**	**Yield[b]**	**Unreacted cyclohexanone[b] (%)**
1.	water	100	81	19
2.	DMF	100	Starting material unconsumed	-
3.	DMSO	140	Starting material unconsumed	-
4.	acetonitrile	100	82	9
5.	isopropanol	80	Starting material unconsumed	-
6.	neat	100	80	11

Reaction conditions: a) cyclohexanone, 20 wt% Mn@CC-SO_3H-NH_2 catalyst, O_2 (20 mL/min), 30 h, b) Isolated yield

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Table 2. Study of synthesis of adipic acid at different temperatures.

Entry	**Temp (°C)**	**Yield[c] (%)**	**Unreacted cyclohexanone[c] (%)**
1.[a]	100	81	19
2.[a]	120	82	7
3.[a]	140	73	12
4.[b]	100	80	11
5.[b]	120	84	8
6.[b]	140	68	10

Reaction conditions: a) cyclohexanone, 20 wt% Mn@CC-SO_3H-NH_2 catalyst, O_2 (20 mL/min), water, 30 h, b) cyclohexanone, 20 wt% Mn@CC-SO_3H-NH_2 catalyst, O_2 (20 mL/min), neat, c) Isolated yield.

Table 3. Study of Mn@CC-SO_3H-NH_2 catalyst concentration for the synthesis of adipic acid.[a]

Entry	**Catalyst conc (wt%)**	**Yield[b] (%)**	**Unreacted cyclohexanone[b] (%)**
1.	5	62	21
2.	10	70	14
3.	15	73	11
4.	20	82	7

Reaction conditions: a) cyclohexanone, O_2 (20 mL/min), water, 120 °C, 30 h, b) Isolated yield.

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Table 4. Comparisons of Mn@CC-SO_3H-NH_2 catalyst *vs*. literature known catalyst for synthesis of adipic acid from cyclohexanone.

Entry	Catalyst	Oxidant	Yield (%)	Selectivity (%)
1.	Hybrid iron phosphonate	O_2	-	96
2.	Titano silicate molecular sieves	O_2	-	94.6
3.	Hollow structured TS-1	O_2	-	95.5
4.	Mn modified Hollow structured TS-1	O_2	-	85.2
5.	Hybrid tin phosphate	None	74	-
6.	Mesoporous triphosphate	None	22	-
7.	H_2WO_4	H_2O_2	91	-
8.	Mn@CC-SO_3H-NH_2	O_2	82	100

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Table 5. Reusability of Mn@CC-SO_3H-NH_2 catalyst in synthesis of adipic acid[a]

Entry	**Yield[b] (%)**	**Unreacted cyclohexanone[b] (%)**
1.	82	11
2.	80	9
3.	81	8
4.	81	9
5.	80	8

Reaction conditions: a) cyclohexanone, 20 wt% Mn@CC-SO_3H-NH_2 catalyst, O_2 (20 mL/min), water, 120 ºC, 30 h. b) Isolated yield.